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Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn-2025

Course Code: CSE-3641 **Course Title: Software Engineering**

Time: 2 hours 30 minutes *Total marks: 50*

[Answer *all* the questions. Figures in the right-hand margin indicate full marks.
Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	DL	Marks
1.a	A hospital information system is being proposed, but doctors and nurses are unsure about the exact features they need. How would requirement elicitation and analysis help in resolving this uncertainty?	2	U	3
1.b	In a library management system, librarians, students, and IT staff provide different requirements. Identify the stakeholders and classify them into primary, secondary, and tertiary stakeholders.	3	U	4
1.c	During validation, several requirements are found to be ambiguous and not testable. Which requirement checking criteria are violated and how can they be corrected.	2	A	3
	OR			
1.a	Identify the primary, secondary, and tertiary stakeholders for the university LMS project.	2	U	3
1.b	Explain the roles and interests of each category of stakeholders in the project.	3	U	4
1.c	Describe how you would engage each group of stakeholders to ensure the project's success.	2	A	3
2.	**Read the passage and answer questions 2(a):** A startup is developing a Campus Food Delivery App for university students. The system includes modules for user registration, restaurant listing, menu browsing, cart management, order placement, payment processing, and delivery tracking. The development team is divided: one group prefers starting with high-level system architecture, while another wants to begin by building and testing individual components first.			
2.a	Compare the Top-Down and Bottom-Up approaches to software design in the context of this project. Explain how each approach would be applied, and state one advantage and one challenge of each.	1	A	5
2.b	A software company based in Bangladesh is currently facing significant challenges in three critical areas: timely and reliable product delivery, efficient utilization and management of physical and digital workspace, and frequent, severe security breaches compromising sensitive data and system integrity. In the context of software engineering requirements, identify and discuss the specific category (or categories) of requirements that address these types of organizational and operational concerns. Clearly define this requirement category, explain its characteristics, and justify why the mentioned issues fall under it rather than other types of software requirements (such as functional or user requirements).	3	E	5

3+4

Group B

Q. No.	Question	CLO	DLE	Marks
3.a	Mr. X, a software developer at a mid-sized tech firm in Dhaka, recently released a pre-final version of a mobile banking application to a group of external end-users as part of a field trial. During this phase, one of the participants entered valid transaction details into the app, but instead of processing the correct amount, the system displayed an incorrect balance deduction. Concerned about reliability and correctness, Mr. X now needs to investigate and resolve this issue before the official launch. **In this context:** I. Briefly define what beta testing is and its role in the software testing lifecycle. II. Identify and explain the general testing approaches or strategies Mr. X should employ to diagnose, isolate, and fix the erroneous behavior reported during beta testing. Justify why these approaches are appropriate for addressing such runtime output discrepancies.	4	E	5
3.b	You are part of a quality assurance team at a software company in Bangladesh developing a web-based e-learning platform that includes features such as user registration, course enrollment, real-time quiz submission, grade calculation, and personalized dashboard analytics. As the testing phase approaches, your team must decide how to allocate testing efforts between black-box and white-box techniques to ensure both functional correctness and internal code reliability. **In this context:** I. Identify and describe specific scenarios or functional features of the e-learning platform that are best suited for black-box testing, and justify why a black-box approach is appropriate for each. II. Similarly, identify and explain specific internal code structures, logic paths, or implementation-level concerns that should be prioritized for white-box testing, along with the rationale for choosing white-box techniques in those cases.	3	A	5
4.a	A well-established fintech startup in Bangladesh has been maintaining a legacy loan-processing system originally developed over a decade ago using outdated technologies and monolithic architecture. Recently, the company has encountered frequent system crashes, difficulty integrating with modern APIs, and an inability to scale during peak user demand. The management has decided not to build a completely new system from scratch due to budget constraints and the risk of losing critical business logic embedded in the legacy code. Instead, they have opted to re-engineer the existing software to improve its structure, performance, and maintainability while preserving its core functionality. **In this context:** I. Elaborate on the step-by-step process of software re-engineering, including key activities such as reverse engineering, code restructuring, data re-engineering, and forward engineering. II. Discuss the major benefits that the fintech company can expect from undertaking software re-engineering, particularly in terms of cost savings, improved reliability, enhanced maintainability, and extended software lifespan.	2	A	5

4.b	The fan in and fan out of module X is 3 and 4 respectively. The complexity of the system is 2000. The number of lines in module X is 200, i.e LOC of module X is LOC(X) =200.Calculate the structural complexity of module X using card and glass's system complexity. Using combined Henry Kafura's approach and Card glass's approach calculate the data complexity of module X.	2	A	5
5.a	What are the key factors that influence software cost estimation? Explain the differences between empirical, heuristic, and analytical estimation techniques.	3	A	4
5.b	Calculate the effort and time required for the three modes (organic, semi-detached, embedded) to complete a 300 KDLOC project using the basic COCOMO model	3	U	6
	OR			
5.	**Read the passage and answer questions 5(a-c):** You are the Director of Quality Assurance at a regional hospital network that has recently adopted an Electronic Health Record (EHR) system. The organization currently operates at CMM Level 2, with consistent project delivery for basic system rollouts. Howèver, there is inconsistency in how clinical workflows are integrated across departments, limited reuse of process assets, and no formalized organizational standards for change management or training. The Chief Medical Officer has mandated a move to CMM Level 3 within 18 months to ensure patient safety, regulatory compliance, and interoperability across all facilities.			
5.a	Outline the core organizational practices you would institutionalize to transition from CMM Level 2 to Level 3, focusing on standardization and documentation.	2	U	3
5.b	Identify two major resistance points you anticipate during this transition (staff, systems, culture), and propose specific mitigation strategies for each.	2	A	4
5.c	Explain how achieving CMM Level 3 will directly improve clinical outcomes and enhance stakeholder (patients, clinicians, administrators) confidence in the EHR system.	2	A	3